

Wilcoxon Signed Ranks test.

KEEL non-parametric statistical module

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1 Detailed results for GWO

1.1 Results

VS	R^+	R^-	Exact P-value	Asymptotic P-value
SA	78606.0	0.0	-	0
HC	78606.0	0.0	-	0

Table 1: Results obtained by the Wilcoxon test for algorithm GWO

1.2 Confidence intervals for Median of differences

$\alpha=0.90$	Confidence interval	Exact confidence
SA	[6,788,193.54 , 6,961,011.135]	2
HC	[5,883,702.96 , 6,002,467.03]	2

Table 2: Confidence intervals for algorithm GWO ($\alpha=0.90$)

$\alpha=0.95$	Confidence interval	Exact confidence
SA	[6,769,125.395 , 6,974,815.235]	2
HC	[5,871,541.59 , 6,014,758.345]	2

Table 3: Confidence intervals for algorithm GWO ($\alpha=0.95$)

2 Detailed results for SA

2.1 Results

VS	R^+	R^-	Exact P-value	Asymptotic P-value
GWO	0.0	78606.0	-	1
HC	219.0	78387.0	-	1

Table 4: Results obtained by the Wilcoxon test for algorithm SA

2.2 Confidence intervals for Median of differences

$\alpha=0.90$	Confidence interval	Exact confidence
GWO	[-6,961,011.135 , -6,788,193.54]	2
HC	[-1,002,729.505 , -979,007.365]	2

Table 5: Confidence intervals for algorithm SA ($\alpha=0.90$)

$\alpha=0.95$	Confidence interval	Exact confidence
GWO	[-6,974,815.235 , -6,769,125.395]	2
HC	[-1,002,802.96 , -978,235.785]	2

Table 6: Confidence intervals for algorithm SA ($\alpha=0.95$)

3 Detailed results for HC

3.1 Results

VS	R^+	R^-	Exact P-value	Asymptotic P-value
GWO	0.0	78606.0	-	1
SA	78387.0	219.0	-	0

Table 7: Results obtained by the Wilcoxon test for algorithm HC

3.2 Confidence intervals for Median of differences

$\alpha=0.90$	Confidence interval	Exact confidence
GWO	[-6,002,467.03 , -5,883,702.96]	2
SA	[979,007.365 , 1,002,729.505]	2

Table 8: Confidence intervals for algorithm HC ($\alpha=0.90$)

$\alpha=0.95$	Confidence interval	Exact confidence
GWO	[-6,014,758.345 , -5,871,541.59]	2
SA	[978,235.785 , 1,002,802.96]	2

Table 9: Confidence intervals for algorithm HC ($\alpha=0.95$)